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NetExplain - Explainable Home & Last-Mile Connectivity Diagnosis

Category: Smart Cities

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Measure. Explain. Resolve.

Problem Description : Today's connectivity problems cannot always be explained by download speed alone.

Key problems

- Traditional speed tests may not reveal packet loss, latency, DNS delay and performance under load.
- Non-technical users cannot easily determine whether a problem is caused by:
- Local gateway/connectivity
- DNS failure
- Packet loss
- Network congestion
- Existing diagnostic commands provide raw measurements but require technical knowledge to interpret.
- Users lack a simple way to understand why a particular problem is suspected.
- ISP support often receives vague complaints such as "Internet is slow" instead of structured diagnostic evidence.

Our challenge :
How can we convert network measurements into an understandable, evidence-based diagnosis without making unsupported claims?

Background

Internet quality is more than bandwidth.

Applications such as:

- Video conferencing
- VoIP
- Online learning
- Cloud applications
- Online gaming

depend on stable connectivity, low latency and reliable packet delivery.

Important measurements

Measurement	Purpose
Gateway reachability	Checks local connectivity
RTT / Latency	Measures response delay
Packet loss	Detects failed packet delivery
DNS resolution time	Checks name-resolution performance
Throughput	Measures data-transfer capacity
Loaded latency	Shows performance under traffic

Our key idea :

A combination of measurements provides better diagnostic evidence than a single speed number.

RFC 2330 provides a framework for defining repeatable IP performance measurements and explicitly emphasizes measurement methodology and uncertainty.

Literature & RFC Review

RFC 2330 — IP Performance Metrics Framework :
Provides the foundation for defining measurable and repeatable IP performance metrics.

Connectivity & Performance Metrics :
IETF IP Performance Measurement work defines approaches for measuring connectivity, delay and packet loss.

Existing diagnostic tools
Tools such as:

- Ping
- Traceroute
- DNS lookup
- Throughput tests

provide useful measurements but generally require users to interpret the results themselves.

Identified gap :
Measurements exist. The missing layer is simple, evidence-based explanation.

Our approach
Probe → Measure → Correlate → Explain → Report

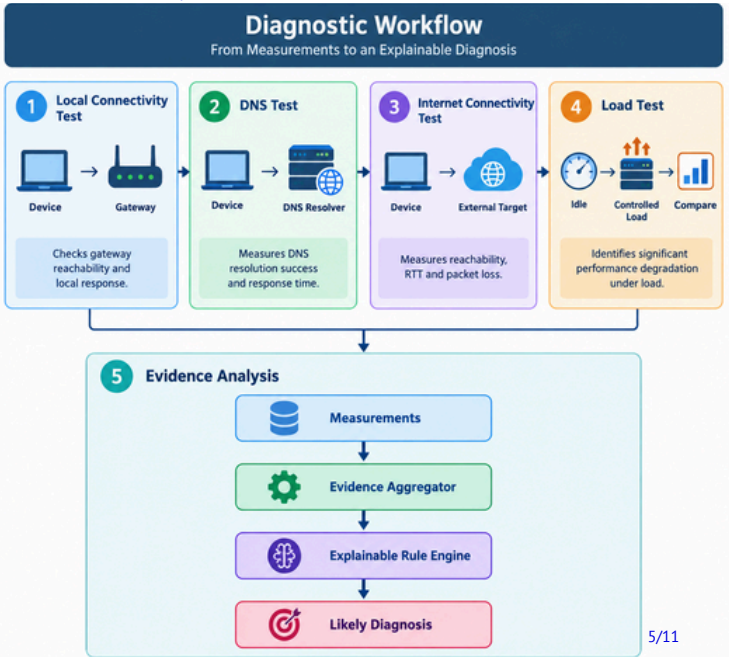
"Our focus is to combine multiple measurements into a guided, explainable diagnosis."

Proposed Solution

NetExplain :
A lightweight **desktop diagnostic agent** that performs guided connectivity tests and converts the results into an explainable diagnosis.

Four controlled lab scenarios :

- Gateway unreachable
- DNS failure
- Induced packet loss
- Network congestion



Optimization Proposed

Our 5 differentiators :

1. Explainability Matrix
Every diagnosis shows the measurements that triggered the rule.

2. Differential Diagnosis
The system compares possible causes instead of blindly choosing one.

3. Inconclusive Safety
If evidence is insufficient, the system says INCONCLUSIVE.

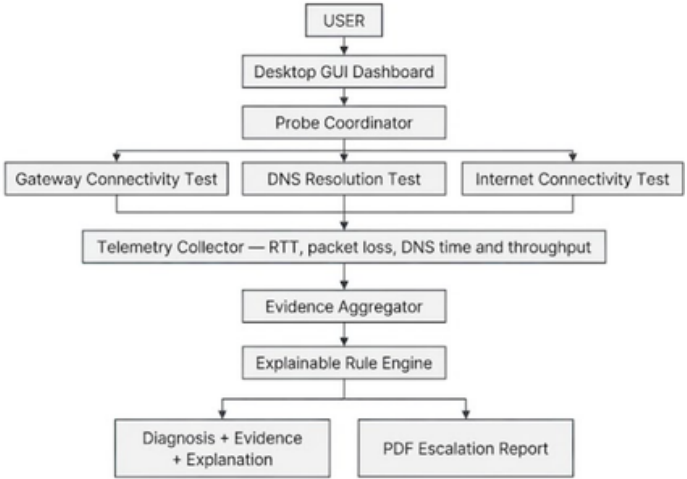
4. Local-vs-Path Isolation
Compares gateway behavior with external-target behavior.

5. One-Click Escalation Report
Creates a structured report containing measurements, diagnosis and evidence.



NetExplain does not just say "your Internet is slow" — it explains what the measurements suggest and why.

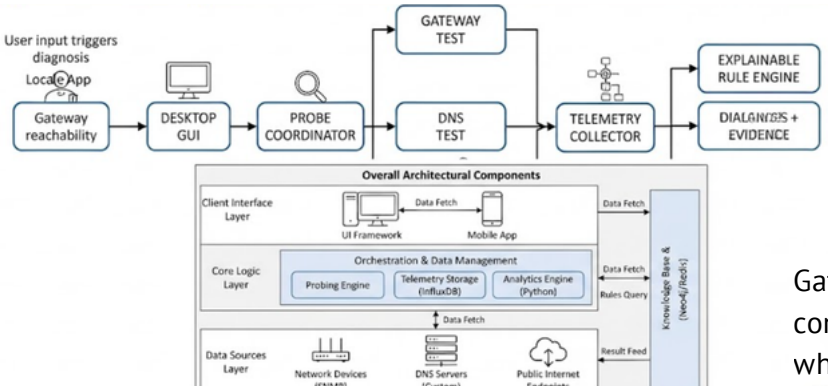
Solution Architecture & Design



Technology

- Python + PySide6 + networking libraries + SQLite
- Controlled lab: Docker/Linux + network fault simulation.
- This is realistic enough for an Antigravity-generated MVP.

Data Flow Diagram



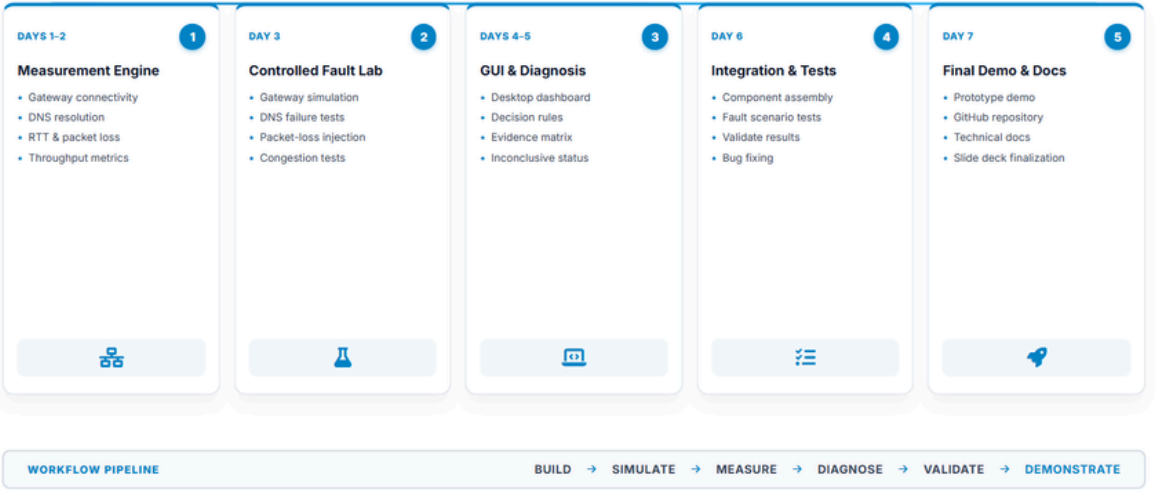
Likely DNS Failure — High Evidence

- Gateway: PASS
- Direct IP: PASS
- DNS: FAIL
- Packet Loss: 0%

Gateway and direct IP connectivity are available, while DNS resolution is failing. The evidence therefore points toward a DNS-related problem.

Deployment Timeline

NetExplain | 7-Day Prototype Development Roadmap



Bibliography I

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Thank You for Your Attention!

Thank You

NetExplain

Measure. Explain. Resolve.

From raw network measurements to understandable diagnosis.

Questions?